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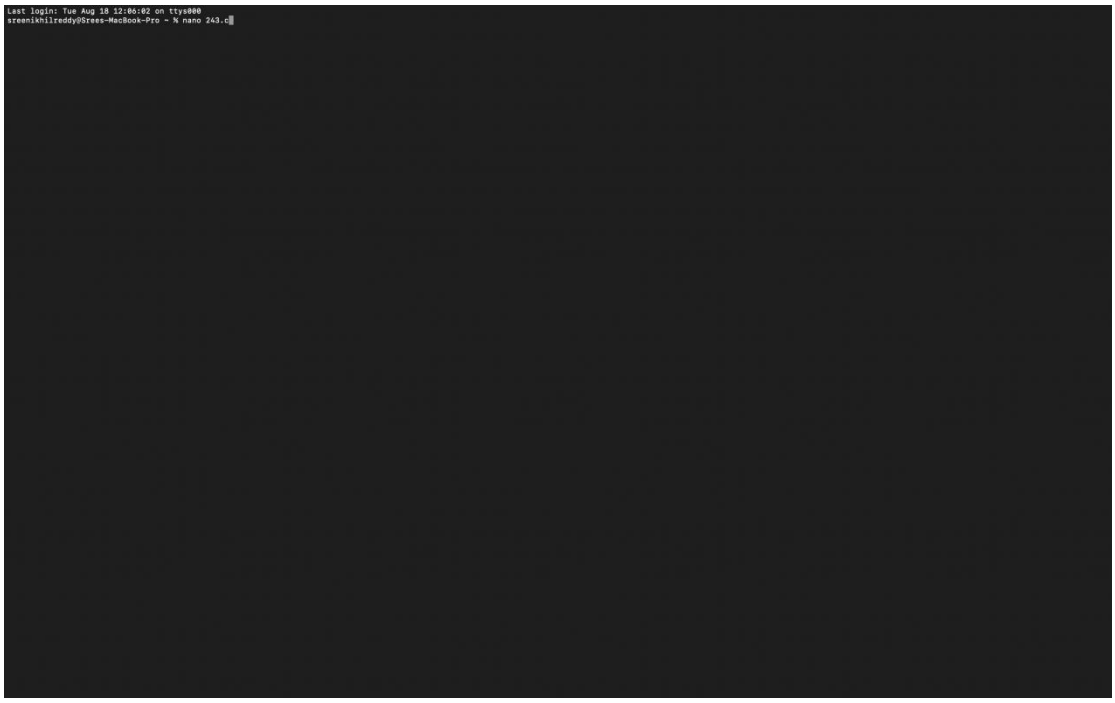
Week 1

SESSION-1

Task 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Make file.



CODE:

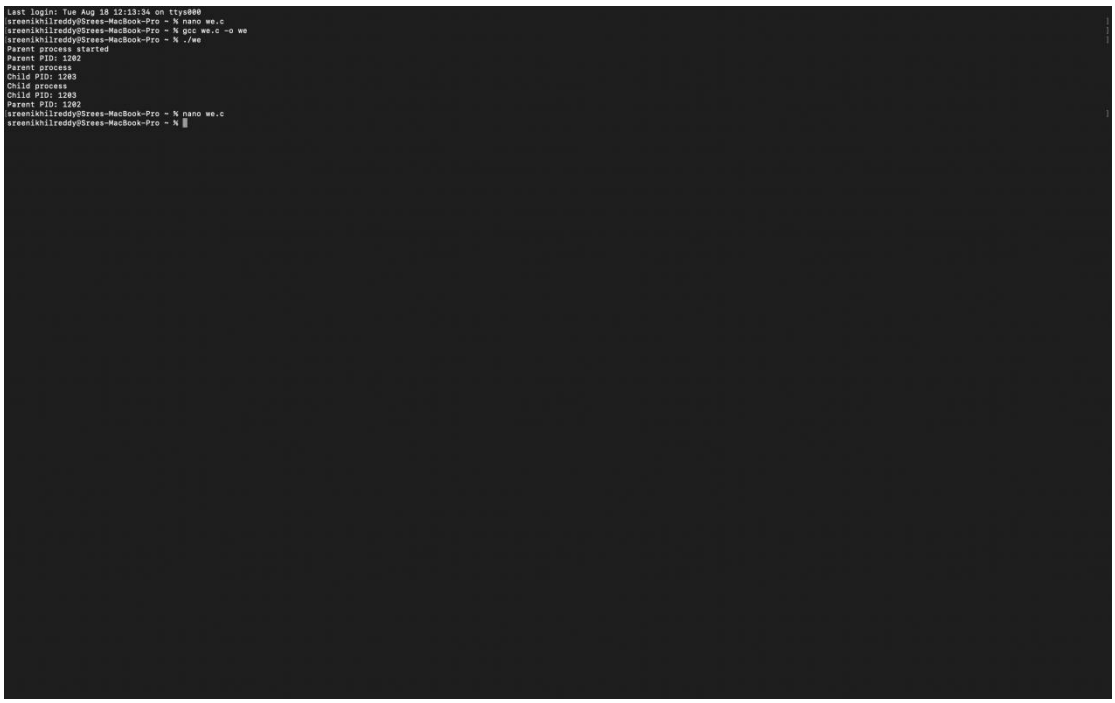
OUTPUT:

```
Last login: Tue Aug 18 12:06:02 on ttys000
sreenikhilreddy@Srees-MacBook-Pro ~ % nano 243.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc 243.c -o 243
sreenikhilreddy@Srees-MacBook-Pro ~ % ./243
Hello, Linux Project!
sreenikhilreddy@Srees-MacBook-Pro ~ %
```

Task 2: To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, practice system call tracing.

```
  UW PICO 5.09                                File: we.c

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        printf("Child process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
    }
    else {
        printf("Parent process\n");
        printf("Child PID: %d\n", pid);
        wait(NULL);
    }

    return 0;
}

^G Get Help  ^O WriteOut  ^R Read File  ^Y Prev Pg  ^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where is  ^V Next Pg  ^U UnCut Text ^T To Spell
```

OUTPUT:

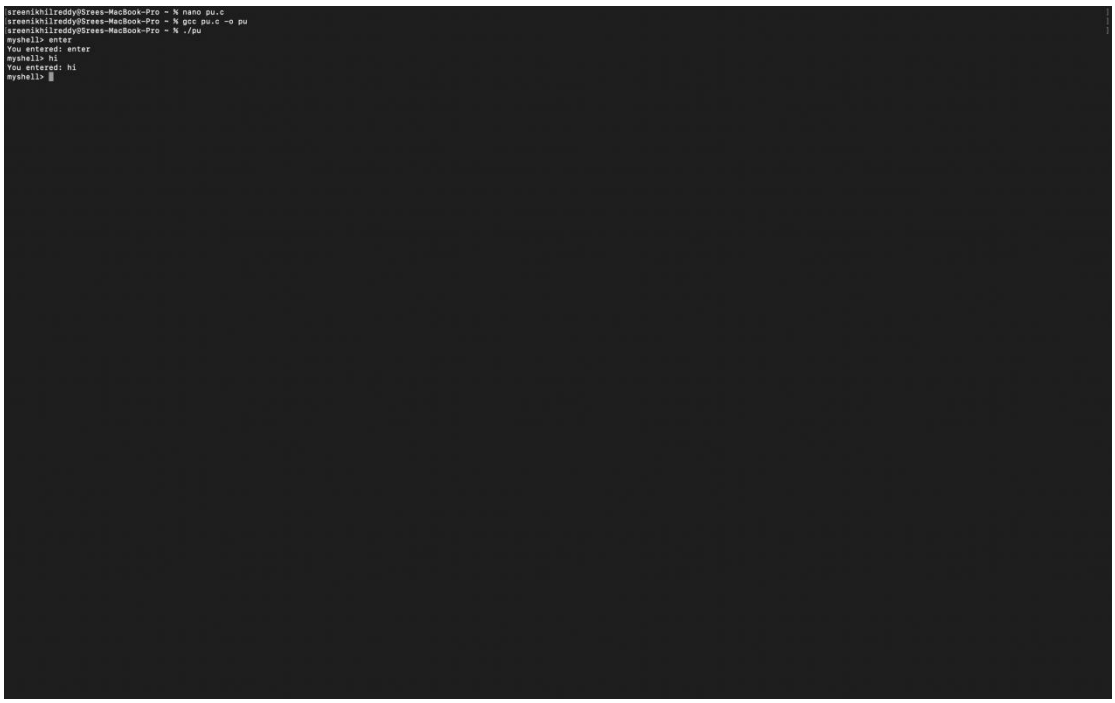


SESSION-2

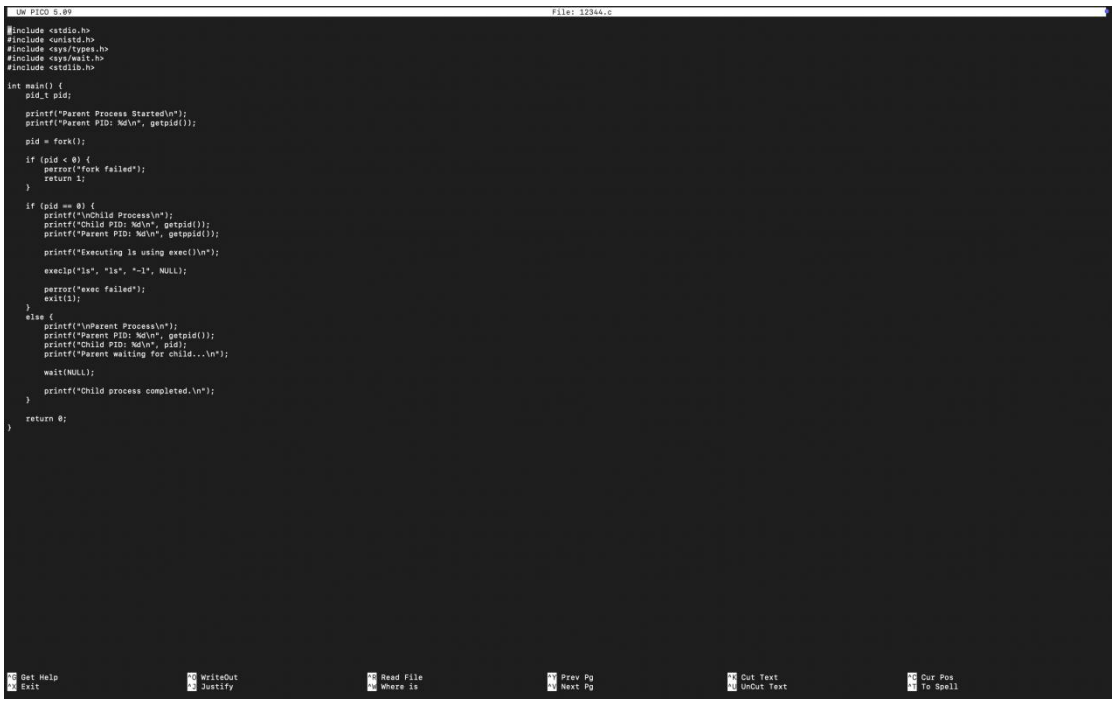
Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

CODE:

OUTPUT:



Task 2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.



OUTPUT:

```
sreenikhilreddy@Srees-MacBook-Pro ~ % nano 12344.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc 12344.c -o 12344
sreenikhilreddy@Srees-MacBook-Pro ~ % ./12344
Parent Process Started
Parent PID: 1389

Parent Process
Parent PID: 1389
Child PID: 1390
Parent waiting for child...

Child Process
Child PID: 1390
Parent PID: 1389
Executing ls using exec()
total 1216
-rw-r--r--    1 sreenikhilreddy  staff   286 15 Aug 11:31 00.c
-rw-------    1 sreenikhilreddy  staff   183 25 Jul 09:24 1.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff  9000 15 Aug 11:00 100
-rw-r--r--    1 sreenikhilreddy  staff  1109 15 Aug 11:02 100.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8776 15 Aug 11:25 101
-rw-r--r--    1 sreenikhilreddy  staff  1241 15 Aug 11:26 101.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8776 15 Aug 11:27 111
-rw-r--r--    1 sreenikhilreddy  staff  1241 15 Aug 11:27 111.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8736 15 Aug 12:03 1111
-rw-r--r--    1 sreenikhilreddy  staff  1018 15 Aug 12:03 1111.c
-rw-------    1 sreenikhilreddy  staff  1019 15 Aug 12:05 1111.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff  8680 25 Jul 09:39 12
-rw-r--r--    1 sreenikhilreddy  staff   183 25 Jul 09:40 12.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8736 11 Aug 12:06 123
-rw-r--r--    1 sreenikhilreddy  staff  1069 13 Aug 13:45 123.c
-rw-------    1 sreenikhilreddy  staff  1069 15 Aug 10:59 123.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff  8528 11 Aug 12:11 1234
-rw-r--r--    1 sreenikhilreddy  staff   357 11 Aug 12:11 1234.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8736 18 Aug 12:31 12344
-rw-r--r--    1 sreenikhilreddy  staff   880 18 Aug 12:31 12344.c
-rw-------    1 sreenikhilreddy  staff   186 15 Aug 12:23 123444.c.save
-rw-------    1 sreenikhilreddy  staff  2706 15 Aug 12:21 123456.c.save
-rw-r--r--    1 sreenikhilreddy  staff  1042 18 Aug 12:25 1234567.c
-rw-r--r--    1 sreenikhilreddy  staff  1042 18 Aug 12:27 18.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8528  1 Aug 08:20 2
-rw-r--r--    1 sreenikhilreddy  staff   616  1 Aug 08:19 2.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8784 15 Aug 12:26 222
-rw-r--r--    1 sreenikhilreddy  staff   918 15 Aug 12:26 222.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8744 18 Aug 12:00 2222
-rw-r--r--    1 sreenikhilreddy  staff   389 18 Aug 12:00 2222.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8448 18 Aug 12:14 243
-rw-r--r--    1 sreenikhilreddy  staff    88 18 Aug 12:14 243.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8776 15 Aug 11:47 264
-rw-r--r--    1 sreenikhilreddy  staff  1450 15 Aug 11:47 264.c
-rwxr-xr-x    1 sreenikhilreddy  staff  9008  1 Aug 09:26 3
-rw-r--r--    1 sreenikhilreddy  staff  1136  1 Aug 08:27 3.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8792 18 Aug 12:28 333
-rw-r--r--    1 sreenikhilreddy  staff   813 18 Aug 12:27 333.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8736  8 Aug 09:41 34
-rw-r--r--    1 sreenikhilreddy  staff   678  8 Aug 09:41 34.c
-rw-------    1 sreenikhilreddy  staff     1 15 Aug 11:38 36.c.save
-rwxr-xr-x    1 sreenikhilreddy  staff  8744  5 Aug 08:58 4
-rw-r--r--    1 sreenikhilreddy  staff   588  4 Aug 12:31 4.c
-rw-------    1 sreenikhilreddy  staff   636  4 Aug 12:25 4.c.save
-rw-r--r--    1 sreenikhilreddy  staff  1148 15 Aug 10:57 45.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8800 15 Aug 12:23 466
-rw-r--r--    1 sreenikhilreddy  staff   730 15 Aug 12:23 466.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8448 15 Aug 11:59 567
-rw-r--r--    1 sreenikhilreddy  staff   127 15 Aug 11:59 567.c
-rw-r--r--    1 sreenikhilreddy  staff  1148 15 Aug 10:56 a.c
-rwxr-xr-x    1 sreenikhilreddy  staff  8528  8 Aug 09:04 a.out
drwx------@   7 sreenikhilreddy  staff   224 11 Aug 11:38 Applications
drwxr-xr-x@   4 sreenikhilreddy  staff   128 31 Jul 09:13 Applications (Parallels)
-rwxr-xr-x    1 sreenikhilreddy  staff  8776 18 Aug 12:04 cc
-rw-r--r--    1 sreenikhilreddy  staff   644 18 Aug 12:03 cc.c
-rw-r--r--    1 sreenikhilreddy  staff  1241 15 Aug 11:26 copy
drwxr-xr-x    3 root             staff    96 14 Aug 14:29 data
drwx------@  11 sreenikhilreddy  staff   352 18 Aug 12:24 Desktop
```